The documentation and process conversion measures necessary to comply with this revision shall be completed by 31 November 1998

INCH-POUND

MIL-PRF-19500/558C
31 August 1998
SUPERSEDING
MIL-S-19500/558B
14 June 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FOUR TRANSISTOR ARRAY,UNITIZED,
PNP, SILICON, SWITCHING TYPES 2N6987, 2N6987U, AND 2N6988,
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors, four independent chip array. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1, 3, and 4 (14-pin dual-in-line, 14-pin flat-pack, and 20-pin leadless chip carrier) and (see 3.3).

1.3 Maximum ratings. 1/

	P_T T_A = +25°C 2/	V_{CBO} 3/	V_{EBO} 3/	V_{CEO} 3/	I_C 3/	T_{OP} and T_{STG}
	W	V dc	V dc	V dc	mA dc	°C
2N6987	1.5	60	5	60	600	-65 to +200
2N6987U	1.0	60	5	60	600	-65 to +200
2N6988	0.4	60	5	60	600	-65 to +200

1/ Maximum voltage between transistors shall be ≥ 500 V dc.
2/ Derate linearly 8.57 mW/°C above T_A = +25°C for 2N6987 and 2N6987U.
Derate linearly 2.286 mW/°C above T_A = +25°C for 2N6988. Ratings apply to total package.
3/ Ratings apply to each transistor in the array.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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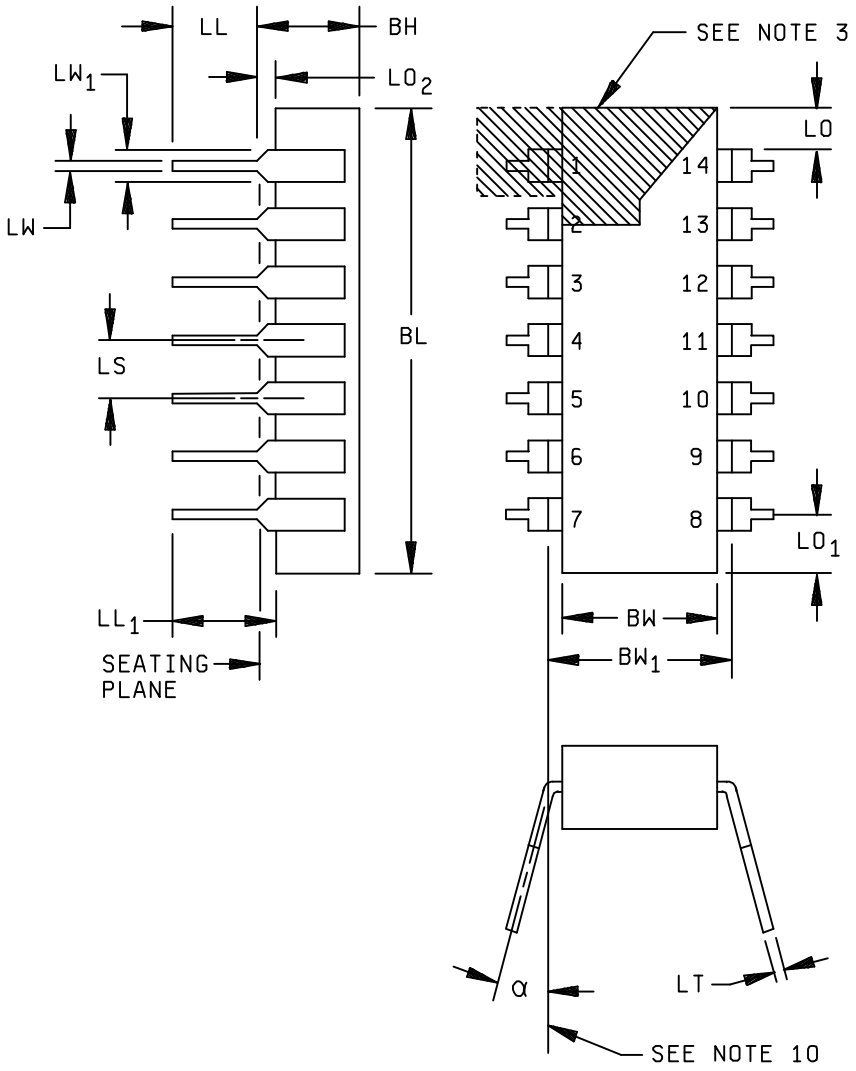


FIGURE 1. Dimensions and configuration for type 2N6987.

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Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
BH	---	.200	---	5.08	
LW	.014	.023	0.36	0.58	8
LW_1	.030	.070	0.76	1.78	4, 8
LT	.008	.015	0.20	0.38	8
BL	---	.785	---	19.94	4
BW	.220	.310	5.59	7.87	4
BW_1	.290	.320	7.37	8.13	7
LS	.100 BSC		2.54 BSC		5, 9
LL	.125	.200	3.18	5.08	
LL_1	.150	---	3.81	---	
LO_2	.015	.060	0.38	1.52	3
LO_1	---	.098	---	2.49	6
LO	.005	---	0.13	---	6
α	0°	15°	0°	15°	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Index area; a notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
4. The minimum limit for dimension LW_1 may be .023 inch (0.58 mm) for lead numbers 1, 7, 8, and 14 only.
5. Dimension LO_2 shall be measured from the seating plane to the base plane.
6. This dimensions allows for off-center lid, meniscus, and glass overrun.
7. The basic pin spacing is .100 inch (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 inch (0.25 mm) of its exact longitudinal position relative to pins 1 and 14.
8. Applies to all four corners (lead numbers 1, 7, 8, and 14).
9. Lead center when α is 0°. BW_1 shall be measured at the centerline of the leads.
10. All leads: Increase maximum limit by .003 inch (0.08 mm) measured at the center of the flat, when lead finish A is applied. Pointed or round lead ends are allowed.
11. Twelve spaces.
12. No organic or polymeric materials shall be molded to the bottom of the package to cover leads.

FIGURE 1. Dimensions and configuration for type 2N6987 - Continued.

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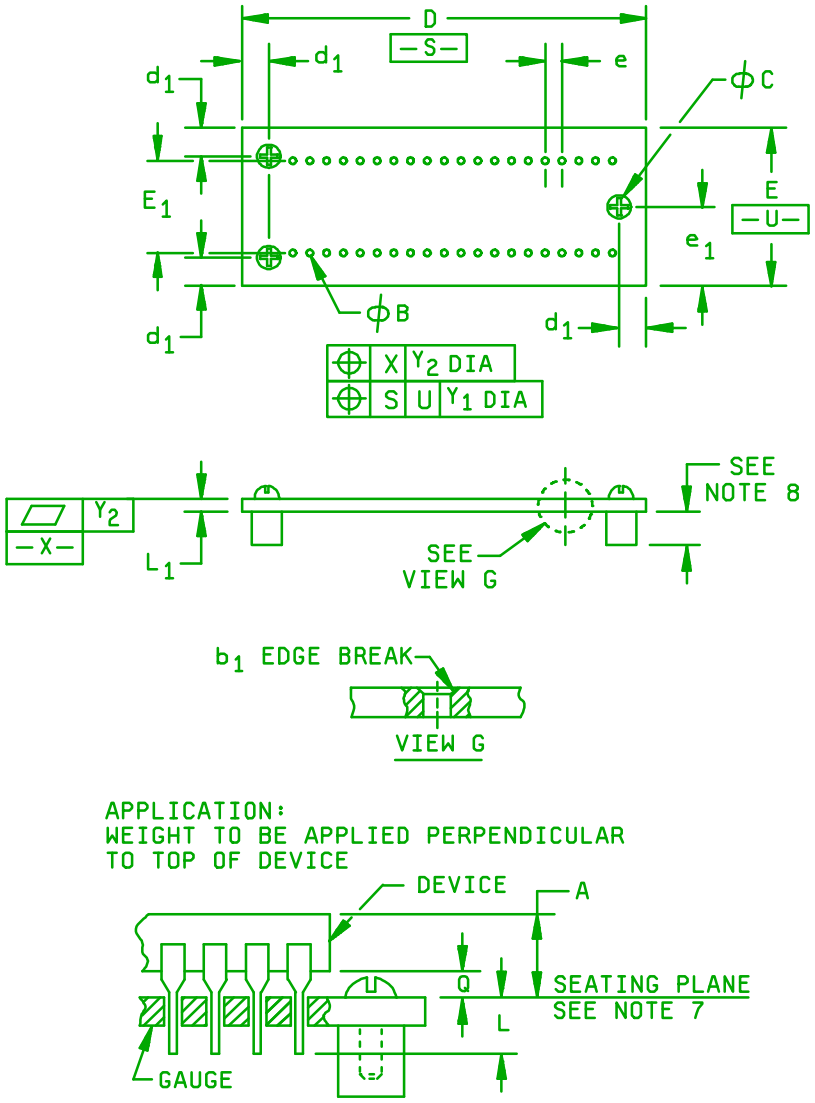


FIGURE 2. Lead gauge.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
∅B	.0323	.0333	.82	.85	
b_1		.002		.05	
∅C	.119	.129	3.02	3.28	4
D	1.750		44.45		
d_1	.200 NOM		5.08 NOM		
E	.750		19.05		4
E_1	.300 BSC		7.62 BSC		
e	.100 BSC		2.54 BSC		
e_1	.375		9.53		6
L_1	.090	.099	2.29	2.51	4
N	14		14		2
Y_1		.010		.25	
Y_2		.002		.05	3, 5
W	8	12	8	12	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensioning and tolerancing per ANSI Y14.5.
4. N is the number of ∅B lead position holes in each gauge variation.
5. W equals the milligrams of weight per lead to be applied perpendicular to the top of devices.
6. Maximum dimensions at D, E, and e_1 are manufacturing options with e_1 = E/2.
7. The seating plane shall be established by inserting the terminal leads of a device into the gauge with weight (W) applied to the top of the device. Dimensions A, Q, and L (see figure 1) can then be measured.
8. Height of leg optional, but sufficient to prevent lead of inserted device from bottoming on measuring table.

FIGURE 2. Lead gauge - Continued.

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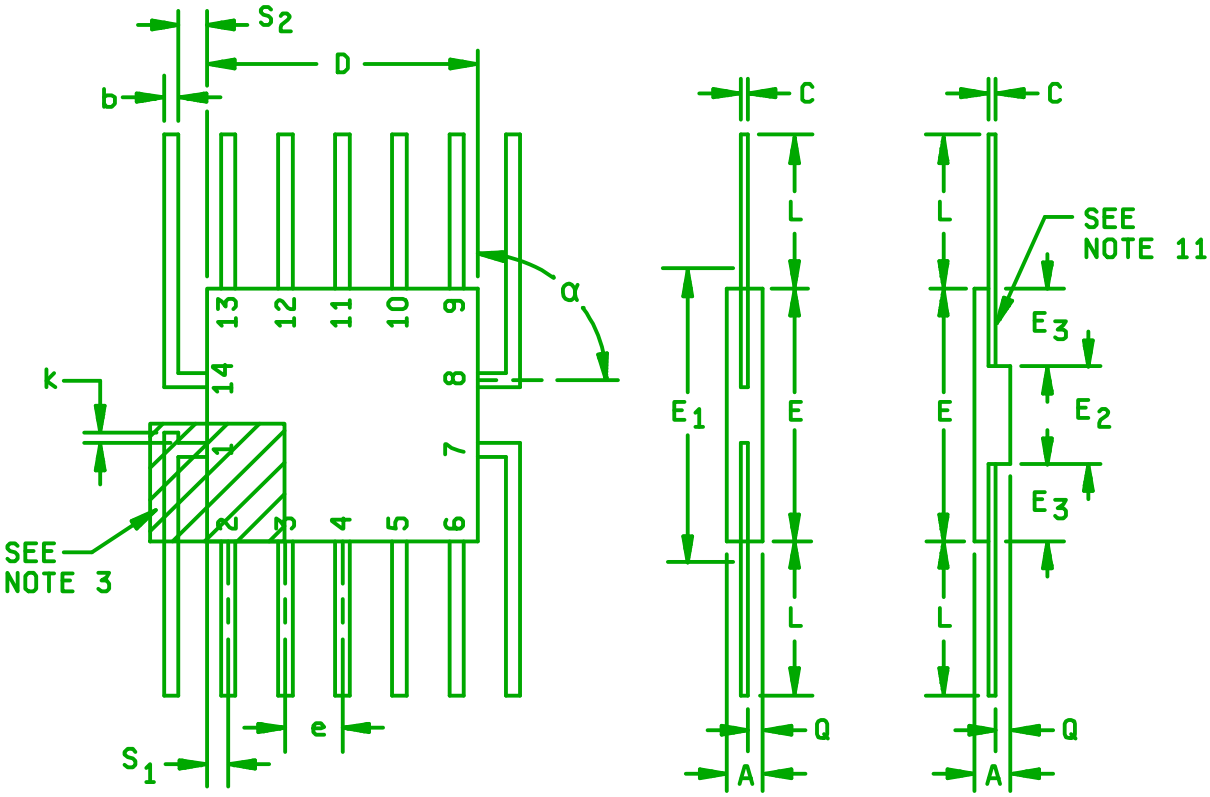


FIGURE 3. Physical dimensions for type 2N6988.

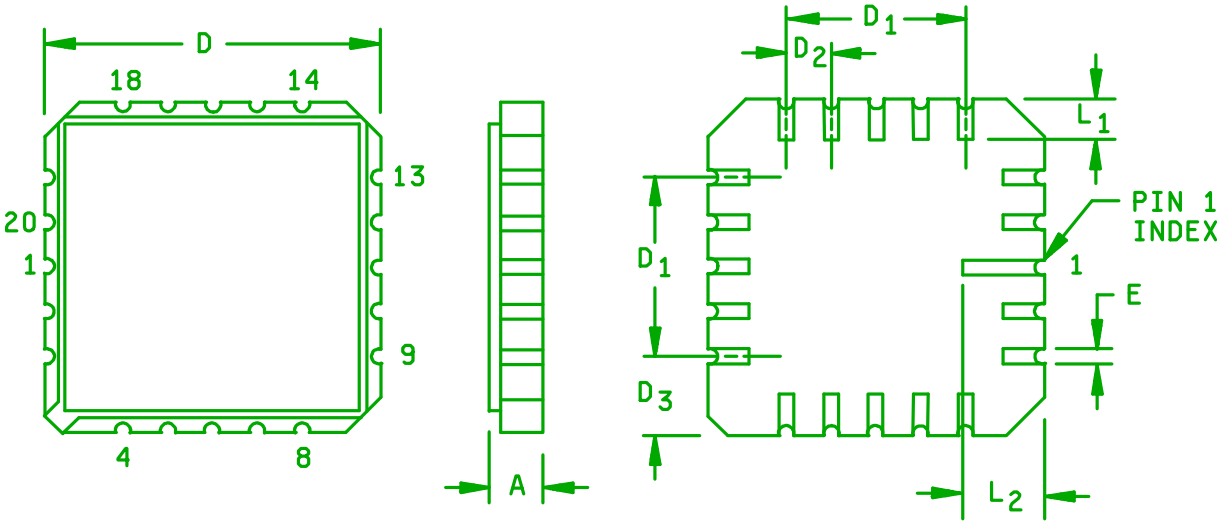
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.030	.115	.76	2.92	
b	.010	.019	.25	.48	5
C	.003	.006	.08	.15	5
D		.280		7.11	3
E	.240	.260	6.10	6.60	
E_1		.290		7.37	3
E_2	.125		3.18		
E_3	.030		.76		
e	050 BSC		1.27 BSC		4, 6
k	.008	.015	.20	.38	10
L	.250	.370	6.35	9.40	
Q	.005	.040	.13	1.02	2
S_1	.005		.13		7, 8
S_2	.004		.10		11
α	30°	90°	30°	90°	12

NOTES:
1. Index area; a notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim k) may be used to identify pin one.
2. Dimension Q shall be measured at the point of exit of the lead from the body.
3. This dimension allows for off-center lid, meniscus and glass overrun.
4. The basic pin spacing is .050 inch (1.25 mm) between centerlines. Each pin centerline shall be located within ± .005 inch (.13 mm) of its exact longitudinal position relative to pins 1 and 14.
5. All leads - increase maximum limit by .003 inch (.08 mm) measured at the center of the flat when the lead finish is solder.
6. Twelve spaces.
7. Applies to all four corners (leads number 2, 6, 9, and 13).
8. Dimension S_1 may be .000 inch (.00 mm if leads number 2, 6, 9, and 13) bend toward the cavity of the package within one lead width from the point of entry of the lead into the body or if the leads are brazed to the metallized ceramic body.
9. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
10. Optional, see note 1. If a pin one identification mark is used in addition to this tab, the minimum limit of dimension k does not apply.
11. Applies to leads number 1, 7, 8, and 14.
12. Lead configuration is optional within dimension E except dimensions b and c apply.
13. Dimensions are in inches.
14. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions for type 2N6988 - Continued.

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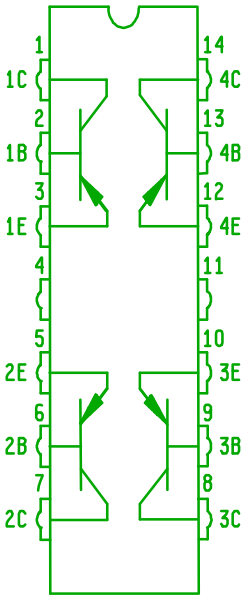


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.063	.075	1.60	1.90
D	.345	.355	8.76	9.02
D_1	.195	.205	4.95	5.21
D_2	.050 TYP		1.27 TYP	
D_3	.070	.080	1.76	2.03
E	.025 REF		0.64 REF	
L_1	.150 REF		3.81 REF	
L_2	.080	.090	2.03	2.28

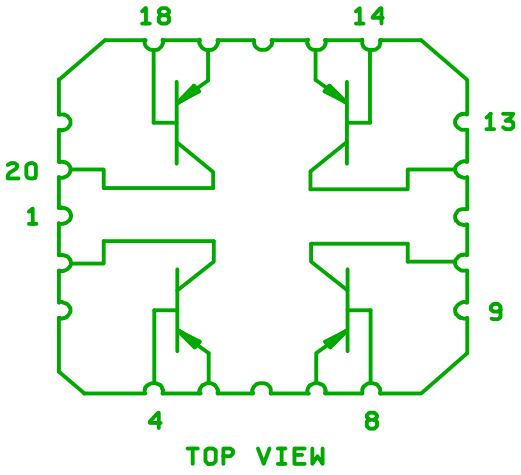
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm).

FIGURE 4. Physical dimensions for type 2N6987U.

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14-lead flat-package or dual-in-line (top view)



20 pin leadless chip carrier (top view).

FIGURE 5. Schematic and terminal connections .

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1.4 Primary electrical characteristics. Characteristics apply to each transistor in the array.

Limits	h_{FE2} 1/ V_{CE} = 10 V dc I_C = 1.0 mA dc	h_{FE4} 1/ V_{CE} = 10 V dc I_C = 150 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	Switching	
				t_{on} see fig.7	t_{off} see fig.8
			pF	ns	ns
Min	100	100			
Max	450	300	8	45	300

Limits	h_{fe} V_{CE} = 20 V dc I_C = 50 mA dc f = 100 MHz	$V_{CE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc
		V dc	V dc
Min	2.0		
Max	8.0	1.6	2.6

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (MO-036AB), 3 and 4 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Schematic and terminal connections. The schematic and terminal connections shall be as shown on figure 5.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and table II herein.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurements	
	JANS level	JANTX & JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CBO2} and h_{FE2}	Not applicable
10	24 hours minimum	24 hours minimum
7(a) or14(a)	Maximum leak rate = 1 x 10^{-7} atm cc/s Test condition C	Maximum leak rate = 1 x 10^{-7} atm cc/s Test condition C
11	I_{CBO2} and h_{FE2} ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent of initial value.	I_{CBO2} and h_{FE4}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, (see 4.5); V_{CB} = 10 - 30 V dc; P_T = 1.5 W for 2N6987; P_T = 1.0 W for 2N6987U, and P_T = 0.4 W for 2N6988.

NOTE: No heat sink or forced air cooling on the devices shall be permitted. Power ratings apply to total package.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current ..... 5 mA.

b. I_H forward heating current ..... 200 mA (minimum).

c. t_H heating time ..... 25 - 30 ms.

d. t_{md} measurement delay time..... 60 µs maximum.

e. V_{CE} collector-emitter voltage ..... 10 V dc

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed per MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and group A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2 herein).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and 4.5.5 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 - 30 V dc; ; P_T = 1.5 W for 2N6987; P_T = 1.0 W for 2N6987U, and P_T = 0.4 W for 2N6988; at T_A = Room ambient as defined in the general requirements of MIL-STD-750, (see 4.5); t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.
B5	1027	V_{CB} = 10 - 30 V dc, T_A = +125°C ±25°C for 96 hours; P_T = 1.5 W for 2N6987; P_T = 1.0 W for 2N6987U, and P_T = 0.4 W for 2N6988. at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot T_J = +275°C.

4.4.2.2 Group B inspection (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 V d, T_J = +150°C minimum. No heat sink or forced- air cooling on devices shall be permitted. n = 45, C = 0.
2	1039	The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, C = 0.
3	1032	High- temperature life (non operating), T_A = +200°C, t = 340 hours, n = 22, C = 0.

1/ Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements.:

a. For JAN, JANTX and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.5 herein.

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4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, not applicable for U designation.
C6	1026	1,000 hours at V_{CB} = 10 -30 V dc; T_J = +150°C minimum. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, not applicable for U designation.
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or requalification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with MIL-STD-750, method 3240, except the output capacitor shall be omitted.

4.5.3 Independent transistor inspections. Inspections shall be performed on each transistor in the array.

4.5.4 Transistor-to-transistor resistance. The leads of each transistor shall be shorted together for this test. The resistance shall be measured between each transistor in the array.

4.5.5 Delta Requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 10 V dc	ΔI_{CB02} 1/	100% of initial value or ± 8 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.1	Δh_{FE2} 1/	± 25% change from initial reading.	

1/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical inspection 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 5/		Group A, subgroup 2				
Bond strength 3/ 5/	2037	Precondition T_A = + 250°C at t = 24 hours or T_A = + 300°C at t = 2 hours n = 11 wires, c = 0				
Subgroup 2						
Collector to base, cutoff current	3036	V_{CB} = 60 V dc	I_{CBO1}		10	µA dc
Emitter to base, cutoff current	3061	V_{BE} = 5 V dc	I_{EBO1}		10	µA dc
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO2}		10	nA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 3.5 V dc	I_{EBO2}		50	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}	75		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}	100	450	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}	100		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc pulsed (see 4.5.1)	h_{FE4}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc pulsed (see 4.5.1)	h_{FE5}	50		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.6	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.3	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.6	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO3}		10	µA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE6}	50		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4					*	*
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}	100		
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 50 mA dc; f = 100 MHz	$\|h_{fe}\|$	2.0	8.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8.0	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 2.0 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz; See 4.5.2	C_{ibo}		30	pF
Turn-on time		See figure 7	t_{on}		45	ns
Turn-off time		See figure 8	t_{off}		300	ns
Transistor-to-transistor resistance		$\|V_{T-T}\|$ = 500 V dc; See 4.5.4	R_{T-T}	10^{10}		ohms
Subgroups 5 and 6						
Not applicable						
Subgroups 7						
Decap internal visual (design verification)	2075	n = 1 device, c = 0				

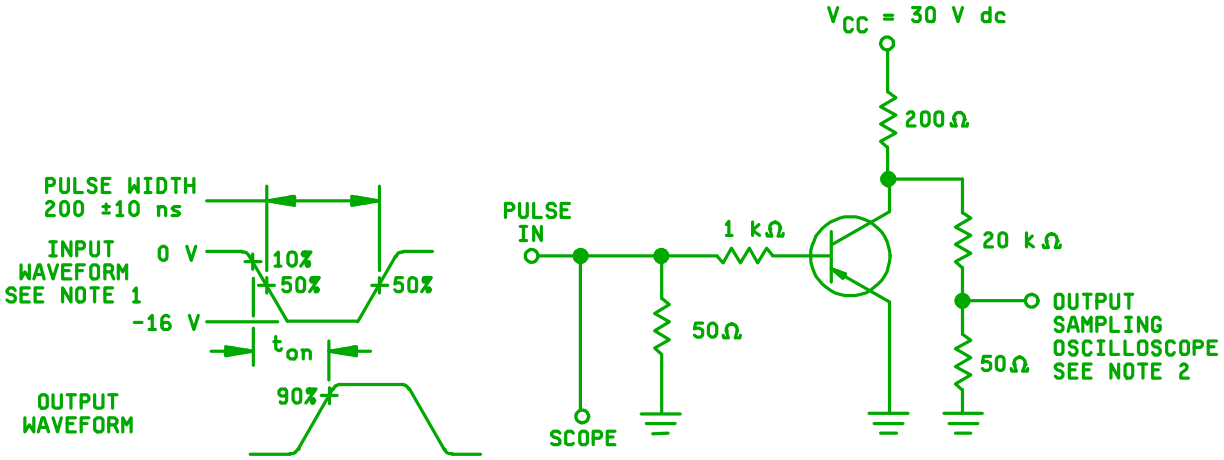
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ Not required for JANS devices.

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TABLE II. Group E inspection (all quality levels) - For qualification only

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 and 4.5.5 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°C$; forced air cooling allowed on cooling cycle only.	
Electrical measurements		See group A, subgroup 2 and 4.5.5 herein.	
Subgroup 3, 4 and 5			
Not applicable			

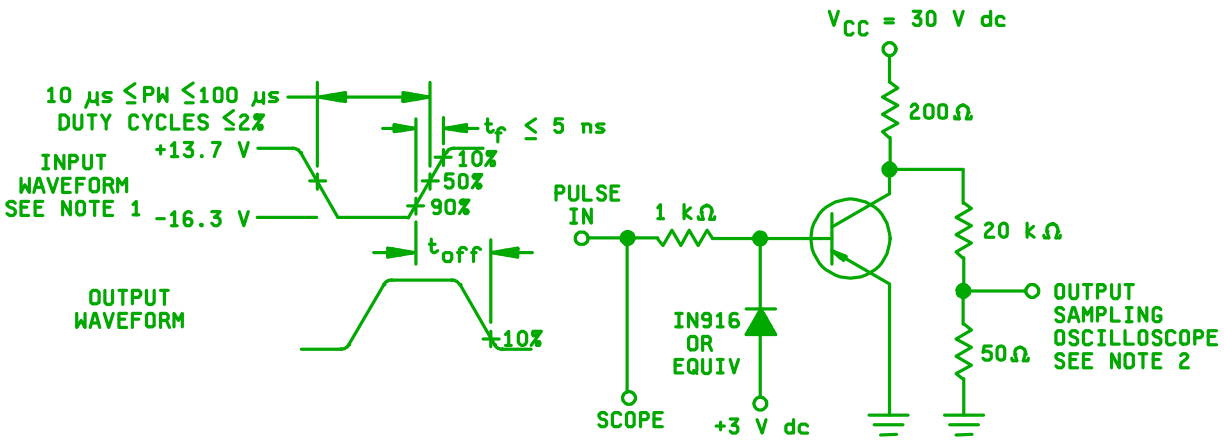
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NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 7. Saturated turn-on switching time test circuit and waveform.



NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 8. Saturated turn-off switching time test circuit and waveform.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - MI, AR, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/558C	**2. DOCUMENT DATE (YYMMDD)** 980831

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, UNITIZED, PNP, SILICON, SWITCHING, FOUR TRANSISTOR ARRAY, TYPES 2N6987, 2N6987U, AND 2N6988, JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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